

A slightly improved upper bound for quantum statistical zero-knowledge

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- 1 Quantum state testing meets quantum statistical zero knowledge (QSZK)
- 2 First result: Algorithmic Holevo–Helstrom measurement and its implication
- 3 Second result: Algorithmic Uhlmann transform and its implication
- 4 Open problems

STATISTICAL DIFFERENCE PROBLEM meets statistical zero-knowledge

Definition 1.1 (Statistical zero-knowledge, informal). An interactive proof system admits the (statistical) zero-knowledge property if verifier's view ($\mathcal{P}_0$) is (statistically) indistinguishable from "verifier's view" ($\mathcal{P}_1$) generated by a polynomial-time simulator.

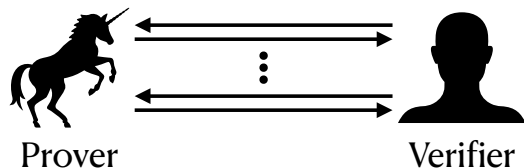


Figure: Verifier's view $\mathcal{P}_0$

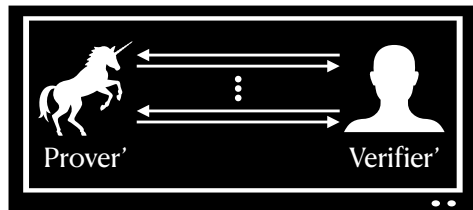


Figure: Simulated "Verifier's view" ($\mathcal{P}_1$)

- ▶ A common misuse (*unrelated to science!*): zero-knowledge vs. zero-entropy.
- ▶ Intuitively, the views $\mathcal{P}_0$ and $\mathcal{P}_1$ can be treated as distributions p_0 and p_1 , respectively. Then *statistical indistinguishability* is captured by the *total variation distance* between p_0 and p_1 .

A complete problem and upper bounds for the class SZK

Definition 1.2 (STATISTICAL DIFFERENCE PROBLEM, $\text{SD}[\alpha, \beta]$) [Sahai–Vadhan'03].

Given efficiently samplable (namely, using polynomial-size Boolean circuits) distributions p_0 and p_1 , decide whether $\text{TV}(p_0, p_1) \geq \alpha$ or $\text{TV}(p_0, p_1) \leq \beta$, where the total variation distance is defined by $\text{TV}(p_0, p_1) := \frac{1}{2} \sum_{x \in S} |p_0(x) - p_1(x)|$.

Specifically, $\text{GAPSD} := \text{SD}[\alpha, \beta]$ where $\alpha(n) - \beta(n) \geq 1/\text{poly}(n)$.

Theorem 1.3 [Sahai–Vadhan'03, Goldreich–Sahai–Vadhan'98].

STATISTICAL DIFFERENCE PROBLEM is SZK-complete. Specifically, $\text{SD}[\alpha, \beta]$ is in SZK if $\alpha^2 - \beta > 0$ for constant α and β .

Upper bounds for SZK and GAPSD

- 1 SZK $\subseteq$ AM [Aiello–Håstad'91, Fortnow'89, Goldreich–Sahai–Vadhan'98].
- 2 SZK = coSZK [Okamoto'00] $\Rightarrow$ SZK $\subseteq$ AM $\cap$ coAM $\subseteq$ PH $\subseteq$ P^{PP}.
- 3 There exists oracle $\mathcal{O}$ such that $\text{SZK}^{\mathcal{O}} \not\subseteq \text{PP}^{\mathcal{O}}$ [Bouland–Chen–Holden–Thaler–Vasudevan'17].
- 4 GAPSD $\in$ AM $\cap$ coAM [Bogdanov–Lee'13].

From quantum ℓ_1 norm to QUANTUM STATE DISTINGUISHABILITY PROBLEM

An n -qubit quantum state ρ is a $2^n \times 2^n$ matrix such that $\text{Tr}(\rho) = 1$ and $\rho \succeq 0$.

Definition 1.4 (QUANTUM STATE DISTINGUISHABILITY PROBLEM, $\text{QSD}[\alpha, \beta]$) [Watrous'02]. Given efficiently preparable (i.e., using polynomial-size quantum circuits) quantum states ρ_0 and ρ_1 , decide whether $T(\rho_0, \rho_1) \geq \alpha$ or $T(\rho_0, \rho_1) \leq \beta$, where the trace distance $T(\rho_0, \rho_1) := \frac{1}{2} \text{Tr}|\rho_0 - \rho_1|$.

Write $\text{GAPQSD} := \text{QSD}[\alpha, \beta]$ when $\alpha(n) - \beta(n) \geq 1/\text{poly}(n)$.

Theorem 1.5 [Watrous'02, Watrous'09]. QUANTUM STATE DISTINGUISHABILITY PROBLEM is QSZK-complete. Specifically, $\text{QSD}[\alpha, \beta]$ is in QSZK if $\alpha^2 - \beta > 0$ for constant α and β .

Upper bounds for QSZK and GAPQSD

- 1 QSZK = $\text{QSZK}_{\text{HV}} \subseteq \text{QIP}(2) \cap \text{co-QIP}(2) \cap \text{PSPACE}$ [Watrous'02, Watrous'09].
- 2 GAPQSD $\in \text{PSPACE}$ [Watrous'02, Le Gall–L.–Wang'23].
- 3 QSZK $\subseteq \text{QIP}(2) \cap \text{co-QIP}(2) \subseteq \text{PSPACE}$ [Jain–Upadhyay–Watrous'09].

An open problem regarding (classical) upper bounds for QSZK [Watrous'02]:

*What further relations among QSZK_{HV} and other complexity classes can be shown? **Almost certainly** the upper-bound of PSPACE on QSZK_{HV} can be improved.*

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Distinguishing quantum states and the Holevo–Helstrom bound

Problem 2.1 (Computational Quantum Hypothesis Testing). Let Q_0 and Q_1 be two polynomial-size quantum circuits acting on n qubits and having r designated output qubits. Let ρ_b denote the quantum state obtained by performing Q_b on the initial state $|0\rangle^{\otimes n}$ and tracing out the non-output qubits for $b \in \{0, 1\}$. Now, consider the following computational task:

- **Input:** A quantum state ρ , either ρ_0 or ρ_1 , is chosen uniformly at random.
- **Output:** A bit b indicates that $\rho = \rho_b$.

Holevo–Helstrom bound

Theorem 2.2 [Holevo'73, Helstrom'69]. Given a quantum state ρ , either ρ_0 or ρ_1 , that is chosen uniformly at random, the maximum success probability to discriminate between quantum states ρ_0 and ρ_1 is given by $\frac{1}{2} + \frac{1}{2}T(\rho_0, \rho_1)$.

An optimal two-outcome measurement $\{\Pi_0, \Pi_1\}$ achieving the maximum discrimination probability:

$$\Pi_0 = \frac{I}{2} + \frac{1}{2}\text{sgn}^{(\text{sv})}\left(\frac{\rho_0 - \rho_1}{2}\right) \text{ and } \Pi_1 = \frac{I}{2} - \frac{1}{2}\text{sgn}^{(\text{sv})}\left(\frac{\rho_0 - \rho_1}{2}\right).$$

It is straightforward to see that $T(\rho_0, \rho_1) = \frac{1}{2}\text{Tr}|\rho_0 - \rho_1| = \text{Tr}(\Pi_0\rho_0) - \text{Tr}(\Pi_0\rho_1)$.

An approximately explicit implementation of the Holevo–Helstrom measurement

Theorem 2.3 (Algorithmic Holevo–Helstrom measurement). Let ρ_0 and ρ_1 be states prepared by n -qubit quantum circuits Q_0 and Q_1 , respectively, as defined in [Problem 2.1](#). An approximate version of the Holevo–Helstrom measurement Π_0 , denoted by $\tilde{\Pi}_0$, can be implemented such that

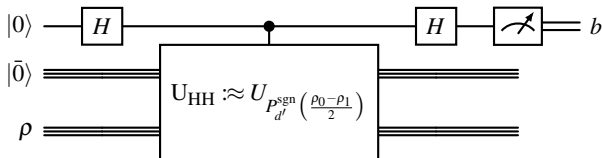
$$|\mathrm{T}(\rho_0, \rho_1) - (\mathrm{Tr}(\tilde{\Pi}_0 \rho_0) - \mathrm{Tr}(\tilde{\Pi}_0 \rho_1))| \leq 2^{-n}.$$

The quantum circuit implementation of $\tilde{\Pi}_0$, acting on $O(n)$ qubits, requires $2^{O(n)}$ uses to the circuits Q_0 , Q_1 , one-, and two-qubit gates. Moreover, the circuit description can be computed in deterministic time $2^{O(n)}$ and space $O(n)$.

Proof Sketch. Instead of implementing $\{\Pi_0, \Pi_1\}$, it suffices to *approximately* implement $\{\hat{\Pi}_0, \hat{\Pi}_1\}$ by the space-efficient QSVT associated with the sign function [\[Le Gall–L.–Wang’23\]](#):

$$\hat{\Pi}_0 = \frac{I}{2} + \frac{1}{2} P_{d'}^{\mathrm{sgn}} \left(\frac{\rho_0 - \rho_1}{2} \right) \text{ and } \hat{\Pi}_1 = \frac{I}{2} - \frac{1}{2} P_{d'}^{\mathrm{sgn}} \left(\frac{\rho_0 - \rho_1}{2} \right).$$

Once we have a block-encoding of $P_{d'}^{\mathrm{sgn}} \left(\frac{\rho_0 - \rho_1}{2} \right)$, we can implement Π_0 :



GAPQSD is in QIP(2) with a quantum linear space honest prover

Theorem 2.4 (GAPQSD is in QIP(2) with a quantum linear-space honest prover). There is a two-message quantum interactive proof system for $\text{QSD}[\alpha(n), \beta(n)]$ with completeness $c(n) = (1 + \alpha(n) - 2^{-n})/2$ and soundness $s(n) = (1 + \beta(n))/2$. Moreover, a *near-optimal* prover strategy for this protocol can be implemented in quantum $2^{O(n)}$ time and $O(n)$ space.

Protocol: Two-message proof system for GAPQSD (quantum linear-space prover).

1. The verifier $\mathcal{V}$ first chooses $b \in \{0, 1\}$ uniformly at random. Subsequently, $\mathcal{V}$ applies Q_b to $|0\rangle^{\otimes n}$, traces out all non-output qubits, and sends the resulting state ρ_b .
2. The verifier $\mathcal{V}$ receives a single-qubit state $\hat{\sigma}$, measures the state in the computational basis, and denotes the outcome by $\hat{b} \in \{0, 1\}$.

The *honest* prover $\mathcal{P}$ measures the received state ρ using the algorithmic Holevo–Helstrom measurement $\{\tilde{\Pi}_0, \tilde{\Pi}_1\}$ (Theorem 2.3), and sends the outcome $\hat{b}$. In particular, the outcome equals $\hat{b}$ if the measurement indicates ρ is $\rho_{\hat{b}}$.

3. The verifier $\mathcal{V}$ accepts if $b = \hat{b}$; otherwise $\mathcal{V}$ rejects.
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Uhlmann fidelity test and the Uhlmann transform

Problem 3.1 (Computational Uhlmann Fidelity Test). Let Q_0 and Q_1 be polynomial-size quantum circuits acting on n qubits in registers (A, R) , each with r output qubits in register A . Let $|\psi_b\rangle$ be the pure state obtained by performing Q_b on $|0^n\rangle$, and let ρ_b be the state obtained by further tracing out the non-output qubits in R for $b \in \{0, 1\}$. Now, consider the following computational task:

- **Input:** The qubits of the purification $|\psi_1\rangle$ in the reference register R , while all qubits in the output register A are fixed and cannot be modified.
- **Output:** A bit z obtained from the measurement $\{\Pi, I - \Pi\}$, where $\Pi := |\psi_0\rangle\langle\psi_0|$.

★ The probability that the Uhlmann fidelity test (Problem 3.1) succeeds (i.e., obtaining the first outcome) is characterized by the (squared) Uhlmann fidelity $F^2(\rho_0, \rho_1) := \text{Tr}(|\sqrt{\rho_0}\sqrt{\rho_1}|)^2$.

Uhlmann's theorem

Theorem 3.2 [Uhlmann'76, Jozsa'94]. Let ρ_0 and ρ_1 be states on register A . For any fixed purification $|\psi_0\rangle$ of ρ_0 on registers (A, R) , the maximum overlap between $|\psi_0\rangle$ and any purification $|\psi_1\rangle$ of ρ_1 on the same registers is given by the fidelity, with $|\psi_1\rangle = U^R|\psi'_1\rangle$:

$$F^2(\rho_0, \rho_1) = \max_{|\psi_1\rangle} |\langle\psi_0|\psi_1\rangle|^2 = \max_U \left| \langle\psi_0|(I^A \otimes U^R)|\psi'_1\rangle \right|^2.$$

An explicit form of the Uhlmann transform

Lemma 3.3 (Explicit form of the Uhlmann transform, [Jozsa'94, Metger–Yuen'23]).

Let $|\psi_0\rangle$ and $|\psi_1\rangle$ be purifications of quantum states ρ_0 and ρ_1 on register A, defined on registers (A, R), where R is the reference register. Let X_{Uhl} be a linear operator on register R. A unitary $U_\star$ on register R that attains the maximum in Uhlmann's theorem (Theorem 3.2) is given by

$$U_\star = \text{sgn}^{(\text{SV})}(X_{\text{Uhl}}), \quad \text{where } X_{\text{Uhl}} := \text{Tr}_A(|\psi_0\rangle\langle\psi_1|^{\text{AR}}).$$

Lemma 3.4 (Exact block-encoding of $\text{Tr}_A(|\psi_0\rangle\langle\psi_1|^{\text{AR}})$, [Utsumi–Nakata–Wang–Takagi'25]).

An *exact* block encoding of X_{Uhl} using n ancillary qubits, implemented by the unitary W on registers (A', R, E), where A' is identical to A and E contains the same number of qubits as R, is given by

$$\langle\bar{0}|^{A'}\langle\bar{0}|^E W |\bar{0}\rangle^{A'}|\bar{0}\rangle^E = X_{\text{Uhl}}, \quad \text{where } W := (Q_1^{A'R})^\dagger (I^{A'} \otimes \text{SWAP}^{R,E}) Q_0^{A'R}.$$

This block-encoding can be implemented using one query to each state-preparation circuit Q_0 and Q_1 , together with $O(n)$ one- and two-qubit quantum gates.

★ When one state is pure, $\text{sgn}^{(\text{SV})}$ is no longer needed in the explicit form of $U_\star$, and thus the exact block-encoding (Lemma 3.4) leads to an optimal estimator for the fidelity [L.–Wang'26].

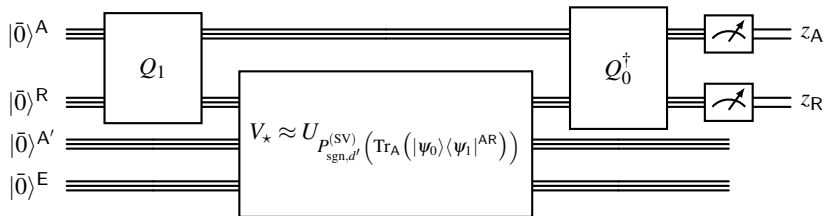
An approximately explicit implementation of the Uhlmann transform

Theorem 3.5 (Algorithmic Uhlmann transform). Let ρ_0 and ρ_1 be quantum states prepared by n -qubit quantum circuits Q_0 and Q_1 , and let $|\psi_0\rangle$ and $|\psi_1\rangle$ denote their purifications before tracing out the non-output qubits, as in [Problem 3.1](#). An approximate version of the Uhlmann transform $U_\star$ specified in [Lemma 3.3](#), denoted by $\tilde{U}_\star$, can be implemented so that

$$F^2(\rho_0, \rho_1) - 2^{-n} \leq \left| \langle \psi_0 |^{\text{AR}} \left(I^{\text{A}} \otimes \tilde{U}_\star^{\text{R}} \right) | \psi_1 \rangle^{\text{AR}} \right|^2 \leq F^2(\rho_0, \rho_1).$$

The quantum circuit implementation of $\tilde{U}_\star$, acting on $O(n)$ qubits, requires $2^{O(n)}$ uses to the quantum circuits Q_0 and Q_1 , as well as $2^{O(n)}$ one- and two-qubit quantum gates. Moreover, the circuit description can be computed in deterministic time $2^{O(n)}$ and space $O(n)$.

Proof Sketch. The quantum circuit construction uses the space-efficient QSVT associated with the sign function [\[Le Gall–L.–Wang’23\]](#) and the exact block-encoding of $\text{Tr}_{\text{A}}(|\psi_0\rangle\langle\psi_1|^{\text{AR}})$:



GAPF²EST is in QIP(2) with a quantum linear space honest prover

Definition 3.6 (QUANTUM SQUARED FIDELITY ESTIMATION PROBLEM, F²EST[α, β]). Given efficiently preparable quantum states ρ_0 and ρ_1 , decide whether $F^2(\rho_0, \rho_1) \geq \alpha$ or $F^2(\rho_0, \rho_1) \leq \beta$.

Write GAPF²EST := F²EST[α, β] when $\alpha(n) - \beta(n) \geq 1/\text{poly}(n)$.

Theorem 3.7 (GAPF²EST is in QIP(2) with a quantum linear-space honest prover). There exists a two-message quantum interactive proof system for F²EST[$\alpha(n), \beta(n)$] with completeness $c(n) = \alpha(n) - 2^{-n}$ and soundness $s(n) = \beta(n)$. Moreover, a *near-optimal* prover strategy for this proof system can be implemented in quantum $2^{O(n)}$ time and $O(n)$ space.

A consequence for QUANTUM STATE CLOSENESS PROBLEM (QSC), the *complement* of QSD:

Corollary 3.8. For any efficiently computable functions $\alpha(n)$ and $\beta(n)$ satisfying $0 \leq \beta(n) < \alpha(n) \leq 1$ and $\alpha^2(n) - \beta(n) \geq 1/O(\log n)$, QSC[β, α] is in QIP(2) with a quantum linear-space honest prover.

GAPF²EST is in QIP(2) with a quantum linear space honest prover (Cont.)

The two-message quantum interactive proof system underlying [Theorem 3.7](#):

Protocol: Two-message proof system for GAPF²EST (quantum linear-space prover).

1. The verifier $\mathcal{V}$ applies Q_0 to $|0\rangle^{\otimes n}$, and sends the *non-output* qubits in register R, while keeping output qubits in register A.
2. The verifier $\mathcal{V}$ receives the (possibly modified) qubits in register R.

The *honest* prover $\mathcal{P}$ applies the algorithmic Uhlmann transform $\tilde{U}_*$ (Theorem 3.5) to the received qubits, and sends the resulting qubits back.

3. The verifier $\mathcal{V}$ applies $Q_1^\dagger$ to the registers (A,R), where register A contains the output qubits of Q_0 and register R contains the received qubits. $\mathcal{V}$ then measures all qubits in (A,R) in the computational basis and accepts if the measurement outcome is the n -bit all-zero string; otherwise, $\mathcal{V}$ rejects.
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Conclusions and open problems

Take-home messages

- 1 The Holevo–Helstrom measurement can be *approximately* implemented by the space-efficient QSVT in quantum $2^{O(n)}$ time and $O(n)$ space. Consequently, GAPQSD is in QIP(2) with a *quantum linear-space* honest prover.
- 2 The Uhlmann transform can be *approximately* implemented by the space-efficient QSVT in quantum $2^{O(n)}$ time and $O(n)$ space. Consequently, GAPF²EST (and thus QSC) is in QIP(2) with a *quantum linear-space* honest prover.

Open problems

- 1 Could the quantum upper bound for QSZK be improved to a subclass of QIP(2) defined via “public-coin” quantum interactive proof systems [Marriott–Watrous’05, Kobayashi–Le Gall–Nishimura’19]?
- 2 Could the classical upper bounds for QSZK be improved to any subclass of PSPACE?
- 3 Could the honest prover’s strategy in any two-message quantum interactive proof system be *approximately* implemented in quantum *linear* space? Namely, the space complexity of the algorithmic implementation scales *linearly* with # of qubits used by the verifier’s first action.

Merci!